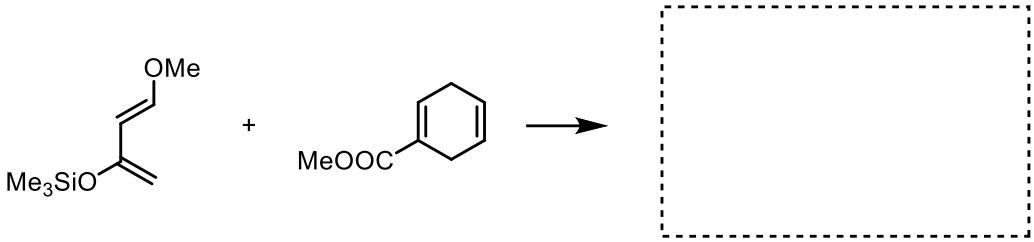
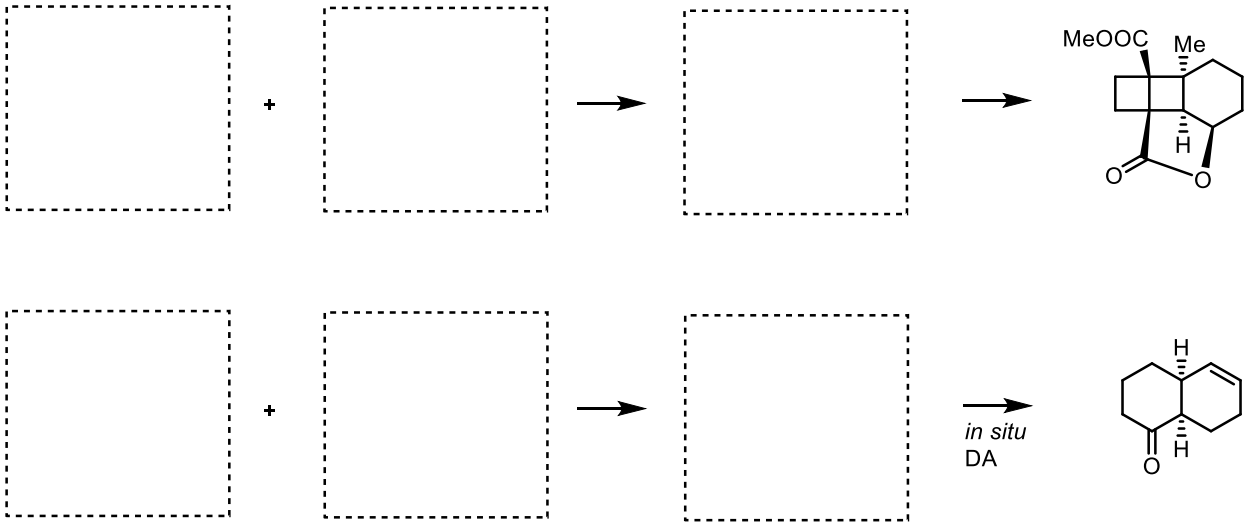
Question 1. Proposed the structures of pericyclic reaction products. Explain the regio- and stereo-selectivity of each reaction if applicable.



Question 2. Propose the reactants and reagents for the synthesis of the target molecules. Explain the regio- and stereo- selectivity of each reaction if applicable.



Question 3. Propose a reaction mechanism for the following reaction (Hint: the key step is a cycloaddition reaction). Explain the regioselectivity. Is this reaction allowed according to Woodward-Hoffmann rules?

